Experiment - 13

Student Name: Aditya Pratap Singh
Branch: CSE - AIML
Semester: 5
Subject Name: ADBMS

UID: 24BAI70438
Section/Group: 24AIT_KRG-1_B
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Aim - To study MongoDB as a NoSQL, document-oriented database and perform database and collection management, CRUD operations, querying, updating, deleting, and grouping/aggregation operations.

Software Used –

- MongoDB / mongosh
- Computer system with MongoDB installed

Theory - MongoDB is a NoSQL database. It is document-based and uses objects in a way similar to JavaScript. Data is organized into databases, collections, and documents.

Commands –

Command	Purpose
show dbs	Show all databases
use db_name	Create/select and use a database
db	Display the current database
db.createCollection("collection_name")	Create a collection
show collections	Show all collections
db.dropDatabase()	Drop the current database
db.collection_name.drop()	Drop a collection

Operations –

insertOne() :

```
car_dealership> db.cars.insertOne({
      maker: "Tata",
      model: "Nexon",
      fuel_type: "Petrol",
      transmission: "Automatic",
      engine: {
          type: "Turbocharged"
      },
      cc: 1199,
      torque: "170 Nm",
      features: [
          "Touchscreen",
          "Reverse Camera",
          "Bluetooth Connectivity"
      ],
      sunroof: false,
      airbags: 2
  })
{
  acknowledged: true,
  insertedId: ObjectId('6ac5d1ca9263d7a0a36e96f3')
}
```

insertMany() :

```
car_dealership> db.cars.insertMany([
    {
        maker: "Hyundai",
        model: "Creta",
        fuel_type: "Diesel",
        transmission: "Manual",
        engine: {
            type: "Turbocharged"
        },
        cc: 1493,
        torque: "250 Nm",
        features: ["Sunroof", "Leather Seats", "Wireless Charging"],
        sunroof: true,
        airbags: 6
    },
    {
        maker: "Maruti Suzuki",
        model: "Baleno",
        fuel_type: "Petrol",
        transmission: "Manual",
        engine: {
            type: "Naturally Aspirated"
        },
        cc: 1197,
        torque: "113 Nm",
        features: ["Touchscreen", "Reverse Camera"],
        sunroof: false,
        airbags: 6
    },
```

```
    {
        maker: "Mahindra",
        model: "XUV500",
        fuel_type: "Diesel",
        transmission: "Manual",
        engine: {
            type: "Turbocharged"
        },
        cc: 2179,
        torque: "360 Nm",
        features: ["Alloy Wheels", "Navigation", "Cruise Control"],
        sunroof: false,
        airbags: 6
    },
    {
        maker: "Honda",
        model: "City",
        fuel_type: "Petrol",
        transmission: "Automatic",
        engine: {
            type: "Naturally Aspirated"
        },
        cc: 1498,
        torque: "145 Nm",
        features: ["Sunroof", "Leather Seats"],
        sunroof: true,
        airbags: 6
    },
    {
        maker: "Tata",
        model: "Harrier",
        fuel_type: "Diesel",
        transmission: "Automatic",
        engine: {
            type: "Turbocharged"
        },
        cc: 1956,
        torque: "350 Nm",
        features: ["Panoramic Sunroof", "Touchscreen"],
        sunroof: true,
        airbags: 6
    }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac5d1e19263d7a0a36e96f4'),
    '1': ObjectId('6ac5d1e19263d7a0a36e96f5'),
    '2': ObjectId('6ac5d1e19263d7a0a36e96f6'),
    '3': ObjectId('6ac5d1e19263d7a0a36e96f7'),
    '4': ObjectId('6ac5d1e19263d7a0a36e96f8')
  }
}
```

find() :

```
car_dealership> db.cars.find()
[
  {
    _id: ObjectId('6ac5d1ca9263d7a0a36e96f3'),
    maker: 'Tata',
    model: 'Nexon',
    fuel_type: 'Petrol',
    transmission: 'Automatic',
    engine: { type: 'Turbocharged' },
    cc: 1199,
    torque: '170 Nm',
    features: [ 'Touchscreen', 'Reverse Camera', 'Bluetooth Connectivity' ],
    sunroof: false,
    airbags: 2
  },
  {
    _id: ObjectId('6ac5d1e19263d7a0a36e96f4'),
    maker: 'Hyundai',
    model: 'Creta',
    fuel_type: 'Diesel',
    transmission: 'Manual',
    engine: { type: 'Turbocharged' },
    cc: 1493,
    torque: '250 Nm',
    features: [ 'Sunroof', 'Leather Seats', 'Wireless Charging' ],
    sunroof: true,
    airbags: 6
  },
  {
    _id: ObjectId('6ac5d1e19263d7a0a36e96f5'),
    maker: 'Maruti Suzuki',
    model: 'Baleno',
    fuel_type: 'Petrol',
    transmission: 'Manual',
    engine: { type: 'Naturally Aspirated' },
    cc: 1197,
    torque: '113 Nm',
    features: [ 'Touchscreen', 'Reverse Camera' ],
    sunroof: false,
    airbags: 6
  },
  {
    _id: ObjectId('6ac5d1e19263d7a0a36e96f6'),
    maker: 'Mahindra',
    model: 'XUV500',
    fuel_type: 'Diesel',
    transmission: 'Manual',
    engine: { type: 'Turbocharged' },
    cc: 2179,
    torque: '360 Nm',
    features: [ 'Alloy Wheels', 'Navigation', 'Cruise Control' ],
    sunroof: false,
    airbags: 6
  },
```

```
[
  {
    _id: ObjectId('6ac5d1e19263d7a0a36e96f7'),
    maker: 'Honda',
    model: 'City',
    fuel_type: 'Petrol',
    transmission: 'Automatic',
    engine: { type: 'Naturally Aspirated' },
    cc: 1498,
    torque: '145 Nm',
    features: [ 'Sunroof', 'Leather Seats' ],
    sunroof: true,
    airbags: 6
  },
  {
    _id: ObjectId('6ac5d1e19263d7a0a36e96f8'),
    maker: 'Tata',
    model: 'Harrier',
    fuel_type: 'Diesel',
    transmission: 'Automatic',
    engine: { type: 'Turbocharged' },
    cc: 1956,
    torque: '350 Nm',
    features: [ 'Panoramic Sunroof', 'Touchscreen' ],
    sunroof: true,
    airbags: 6
  }
]
```

findOne() :

```
car_dealership> db.cars.findOne()
{
  _id: ObjectId('6ac5d1ca9263d7a0a36e96f3'),
  maker: 'Tata',
  model: 'Nexon',
  fuel_type: 'Petrol',
  transmission: 'Automatic',
  engine: { type: 'Turbocharged' },
  cc: 1199,
  torque: '170 Nm',
  features: [ 'Touchscreen', 'Reverse Camera', 'Bluetooth Connectivity' ],
  sunroof: false,
  airbags: 2
}
```

updateOne() :

```
car_dealership> db.cars.updateOne(
|     {model: "Nexon"},
|     {$push: {features: "Heated Seats"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

updateMany() :

```
car_dealership> db.cars.updateMany(
|     {fuel_type: "Diesel"},
|     {$set: {alloys: "yes"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

deleteOne() :

```
car_dealership> db.cars.deleteOne({fuel_type: "Petrol"})
{ acknowledged: true, deletedCount: 1 }
```

deleteMany() :

```
car_dealership> db.cars.deleteMany({fuel_type: "Petrol"})
{ acknowledged: true, deletedCount: 2 }
car_dealership> db.cars.deleteMany({})
{ acknowledged: true, deletedCount: 3 }
```